MAT092: Remedial Course in Mathematics

Emon Hossain[1]

[1]Lecturer
MNS department
Brac University

LECTURE-04

Examples: Injective, Surjective, and Bijective Functions

Let A and B be finite sets. We'll explore all possible behaviors:

Example 1: $A = \{a, b\}$, $B = \{1, 2\}$

Function	$f(a)$	$f(b)$	Injective?	Surjective?	Bijective?
f_1	1	1			
f_2	2	2			
f_3	1	2			
f_4	2	1			

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Example 2: $A = \{a, b\}$, $B = \{1, 2, 3\}$ ($|A| < |B|$)

$f(a) = 1,\ f(b) = 2$: injective but not surjective.

$g(a) = 1,\ g(b) = 1$: neither injective nor surjective.

No surjective functions possible since $|A| < |B|$.

continued...

Example 3: $A = \{a, b, c\}$, $B = \{1, 2\}$ $(|A| > |B|)$

continued...

Example 3: $A = \{a, b, c\}$, $B = \{1, 2\}$ ($|A| > |B|$)

$h(a) = 1,\ h(b) = 2,\ h(c) = 1$: surjective but not injective.

$k(a) = 1,\ k(b) = 1,\ k(c) = 1$: neither injective nor surjective.

No injective functions possible since $|A| > |B|$.

continued...

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$h(a) = 1,\ h(b) = 2,\ h(c) = 1$: surjective but not injective.

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Example 4: $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ ($|A| = |B|$)

continued...

Example 3: $A = \{a, b, c\}$, $B = \{1, 2\}$ ($|A| > |B|$)

$h(a) = 1,\ h(b) = 2,\ h(c) = 1$: surjective but not injective.

$k(a) = 1,\ k(b) = 1,\ k(c) = 1$: neither injective nor surjective.

No injective functions possible since $|A| > |B|$.

Example 4: $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ ($|A| = |B|$)

$p(a) = 1,\ p(b) = 2,\ p(c) = 3$: bijective.

$q(a) = 1,\ q(b) = 1,\ q(c) = 2$: neither injective nor surjective.

Composition Function

$f = \{(1,1),(2,3),(3,1),(4,2)\}$, and $g = \{(1,2),(2,3),(3,1),(4,2)\}$, then $g \circ f = \{(1,2),(2,1),(3,2),(4,3)\}$

Composition of Functions: Definition & Basics

Given $f : A \to B$ and $g : B \to C$, the composition is

$$(g \circ f)(x) = g\big(f(x)\big), \quad x \in A, \quad \text{so } g \circ f : A \to C.$$

Associativity: $h \circ (g \circ f) = (h \circ g) \circ f$ (types match).

Identities: $\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$.

Non-commutativity (in general): $g \circ f \neq f \circ g$.

Typing rule: codomain(f) must fit domain(g).

Non-commutativity & Domain Subtleties

Example (non-commutative): $f(x) = x^2$, $g(x) = x + 1$ on $\mathbb{R}$.

$$(g \circ f)(x) = x^2 + 1, \qquad (f \circ g)(x) = (x+1)^2 = x^2 + 2x + 1.$$

They differ, so $g \circ f \neq f \circ g$.

Inverse caution: $f : \mathbb{R} \to [0, \infty)$, $f(x) = x^2$; $g : [0, \infty) \to \mathbb{R}$, $g(y) = \sqrt{y}$.

$$(g \circ f)(x) = |x| \neq x \text{ for } x < 0.$$

But restricting to $[0, \infty)$ gives $g \circ f = \mathrm{id}_{[0,\infty)}$.

Injectivity & Surjectivity via Composition

Facts:

If $g \circ f$ is injective, then f is injective (but g need not be).

If $g \circ f$ is surjective, then g is surjective (but f need not be).

If f, g bijective, then $g \circ f$ bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Cancellation:

g injective and $g \circ f_1 = g \circ f_2 \Rightarrow f_1 = f_2$.

f surjective and $g_1 \circ f = g_2 \circ f \Rightarrow g_1 = g_2$.

Counterexample: $g \notin \text{inj}$, yet $g \circ f$ inj

Sets: $A = \{1, 2\}$, $B = \{a, b, c\}$, $C = \{X, Y\}$.
Define $g : B \to C$: $g(a) = X$, $g(b) = X$, $g(c) = Y$ (not injective). Define $f : A \to B$: $f(1) = a$, $f(2) = c$.
Then $(g \circ f)(1) = X$, $(g \circ f)(2) = Y$ is injective.

Takeaway: $g \circ f$ injective does not force g injective.

Counterexample: $f \notin$ surj, yet $g \circ f$ surj

Sets: $A = \{1, 2\}$, $B = \{a, b\}$, $C = \{X\}$.
Let $g : B \to C$ be constant: $g(a) = g(b) = X$ (surjective onto C). Let $f : A \to B$ be $1 \mapsto a$, $2 \mapsto a$ (not surjective: misses b).
Then $(g \circ f)$ is surjective onto C.

Takeaway: $g \circ f$ surjective does not force f surjective.